

Phase 2: Data Collection and Preprocessing Phase

2.1 Introduction

Data plays a crucial role in any smart system. In the Smart Waste Management System, data is required to monitor garbage levels, predict future waste accumulation, and optimize collection routes.

Since physical IoT sensors are not available in this project, **simulated data** is generated to mimic real-world sensor behavior. This phase focuses on collecting, storing, and preprocessing that data for further use in the system.

2.2 Data Requirements

The system requires the following types of data:

2.2.1 Garbage Fill Level

- Represents how full the bin is
- Range: 0% to 100%
- Used for:
 - Alert generation
 - Route optimization

2.2.2 Garbage Weight

- Indicates the weight of waste inside the bin
- Helps in:
 - Load estimation
 - Vehicle capacity planning

2.2.3 Bin Status

- Indicates whether the bin is:
 - Empty
 - Partially filled
 - Full

2.2.4 Timestamp Data

- Records when data is generated
 - Used for:
 - Historical analysis
 - Prediction models
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2.3 Data Collection Method

2.3.1 Simulated Sensor Data

Due to lack of hardware, the system uses a **sensor simulation module**.

File Used:

- sensors/simulator.py

Functionality:

- Generates random values at regular intervals
- Mimics real IoT sensor behavior

Example:

- Fill Level: 75%
 - Weight: 12.5 kg
 - Status: Almost Full
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2.3.2 Data Generation Technique

The system uses:

- Python random module
- Predefined thresholds

Logic:

- Random values generated between 0–100
 - Weight calculated based on fill level
 - Status derived from thresholds
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2.4 Data Storage

2.4.1 Database Used

- SQLite Database

2.4.2 Database File

- Created via: database/db_setup.py
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2.4.3 Data Tables

Bin Table

- Bin ID
- Location (optional)
- Status

Sensor Data Table

- Bin ID
 - Fill Level
 - Weight
 - Timestamp
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2.4.4 Data Storage Process

1. Sensor simulator generates data
 2. Data sent to backend (Flask)
 3. Backend stores data in SQLite
 4. Data becomes available for:
 - Dashboard
 - Prediction
 - Optimization
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2.5 Data Preprocessing

Raw data cannot be directly used for analysis. It must be cleaned and processed.

2.5.1 Data Cleaning

- Removes invalid values
 - Handles missing data
 - Ensures consistency
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2.5.2 Data Transformation

- Converts raw values into meaningful formats
 - Example:
 - Fill Level → Percentage
 - Weight → Numeric format
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2.5.3 Data Normalization

- Scales data for better model performance
- Ensures:

- Consistency
 - Accuracy
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2.5.4 Data Structuring

- Organizes data into tables
 - Makes it usable for:
 - Machine learning
 - Visualization
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2.6 Data Flow

Step-by-Step Flow:

1. Random data generated
 2. Sent to backend API
 3. Stored in database
 4. Processed for:
 - Dashboard display
 - Route planning
 - Prediction model
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2.7 Data Validation

To ensure accuracy, the system performs:

- Range checks (0–100%)
 - Type checks (numeric values)
 - Logical checks (weight vs fill level)
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2.8 Challenges in Data Collection

- No real sensor data
 - Random data may not reflect real-world patterns
 - Data inconsistency issues
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2.9 Solutions Implemented

- Used realistic random ranges

- Applied preprocessing techniques
 - Structured data properly for models
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2.10 Importance of Preprocessing

Data preprocessing ensures:

- Accuracy in predictions
- Reliable system behavior
- Better visualization

Without preprocessing, the system may:

- Generate incorrect alerts
 - Provide wrong predictions
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2.11 Advantages of Simulated Data

- Easy to implement
 - No hardware cost
 - Flexible testing
 - Helps in prototype development
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2.12 Limitations

- Not fully realistic
 - Cannot capture real-world variations
 - Limited accuracy in prediction
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2.13 Future Improvements

- Replace simulated data with real IoT sensors
 - Use advanced preprocessing techniques
 - Integrate real-time streaming data
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2.14 Conclusion

The Data Collection and Preprocessing Phase is essential for building a reliable smart waste management system. Even though the system uses simulated data, proper preprocessing ensures that the data is structured, clean, and ready for analysis.

This phase lays the foundation for model development, optimization, and intelligent decision-making in later stages.